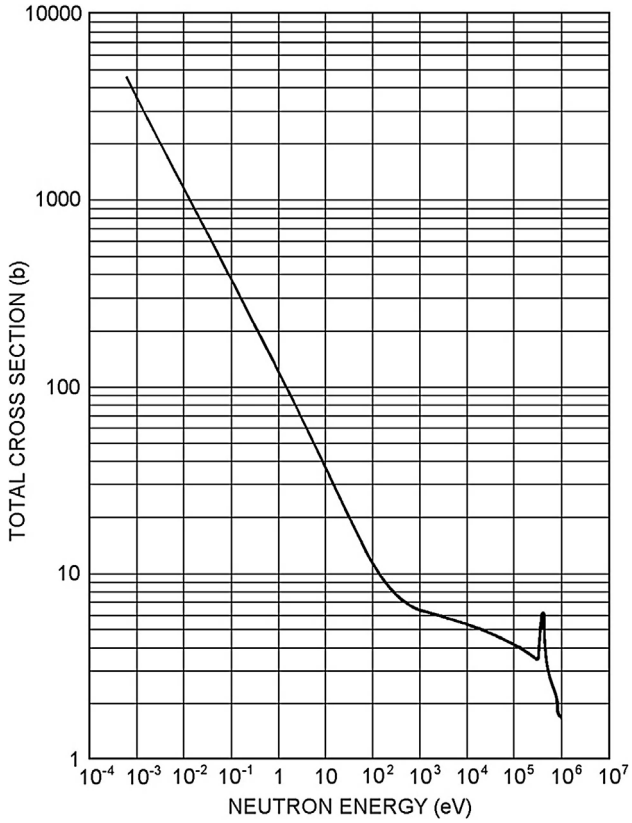


Figure 11.4 The total reaction cross-section of ^{10}B for neutrons of various kinetic energies.

absorption occurs. In Figure 11.4 $\sigma_{tot} = \sigma_{capt} + \sigma_{scat}$, but σ_{scat} is approximately constant. Since σ_{capt} decreases as E_n increases, at higher energies ($\geq$ 500 eV) σ_{scat} dominates over σ_{capt}, except for the resonance at 0.3 MeV. The capture or neutrons in ^{10}B leads to the formation of ^{7}Li and ^{4}He.

11.5. NEUTRON DIFFRACTION

From (11.12) we calculated that for neutrons the wavelength at thermal energies is on the order of 0.1 nm, i.e. of the same order of magnitude as the distance between atomic planes in a crystal. Thermal neutrons can, therefore, be scattered by crystals in the same manner as X-rays. For studies of crystal structures by neutron diffraction, a beam of mono-energetic neutrons can be obtained from the spectrum of neutron energies in a reactor by the use of monochromators.[1]

[1] Another method is mechanical time-of-flight based selectors.

As described in §11.4 neutrons interact with the nucleons in the nucleus. On the other hand photons mainly interact with the electrons surrounding the nucleus (which number is proportional to the atom number) The probability of scattering of neutrons without energy change (*coherent scattering*) is approximately proportional to the area of the nucleus. As a consequence, coherent scattering of neutrons is less dependent on the atomic number than the scattering of X-rays which is proportional to the electron density (i.e. $\propto Z^2$). Thus it is clear that for a compound consisting of both heavy and light atoms, the position of the lighter atoms can be more easily determined using neutron diffraction, while X-ray diffraction is better for locating the heavier atoms. Neutron diffraction is, therefore, valuable for complementing the information obtained on the position of heavy atoms by X-ray diffraction. Neutron diffraction is particularly valuable in the location of hydrogen atoms in organic and biological materials.

11.6. MODELS FOR NUCLEAR REACTIONS

No single model is successful in explaining all the aspects of the various types of nuclear reactions. Let us consider three models which have been proposed for explaining the results of nuclear reaction studies.

11.6.1. The optical model

In the process of elastic scattering the direction of the particles is changed but none of the kinetic energy is converted to nuclear excitation energy. This would indicate that the reaction is independent of the internal structure of the nucleus and behaves much like the scattering of light from a crystal ball. Consequently, a model has been developed based on the mathematical techniques used in optics. Light shining on a transparent crystal ball is transmitted with some scattering and reflection but no absorption. Light shining on a black crystal ball is all absorbed and there is no transmission or scattering. In nuclear reactions the incoming particles are scattered in elastic scattering and are absorbed in induced transmutations. Therefore, if the nucleus is to act as a crystal ball it can be neither totally transparent nor totally black. The optical model of the nucleus is also known as the *cloudy crystal ball model*, indicating that nuclei both scatter and absorb the incoming particles.

The nucleus is described as a potential well containing neutrons and protons. The equation for the nuclear potential includes terms for absorption and scattering. This potential can be used to calculate the probability for scattering of incident particles and the angular distribution of the scattering. The model is in excellent agreement with experiments for scattering. Unfortunately, this model does not allow us to obtain much information about the consequences of the absorption of the particles which lead to inelastic scattering and transmutation.